



New Records of *Cirrhipathes* (Order Antipatharia) for Gulf of Mannar Marine Biosphere Reserve, India

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Abstract

The present study dealt with the description and distribution of two species viz. *Cirrhipathes spiralis*, *Cirrhipathes anguina* under the subclass Hexacorallia, order Antipatharia reported from the Gulf of Mannar Biosphere Reserve, India. Earlier *Cirrhipathes anguina* was reported from Andaman and Nicobar Islands in 1903, and *Cirrhipathes spiralis* was reported from Galle, Sri Lanka coast. Both of the species *C. spiralis* and *C. anguina* are new distribution for the Gulf of Mannar, Tamil Nadu coast of India.

Significance Statement

The MS describes two newly recorded black corals from a protected marine habitat of Tamil Nadu, India - Gulf of Mannar Marine Biosphere Reserve. Such work from a marine biosphere reserve as well as on Antipatharia is very less and to protect the coastal ecosystem of India, cataloguing the indigenous diversity from the region is essential.

Keywords Antipatharian · Black coral · Gulf of Mannar · Horny coral · Thorny coral · India

Antipatharians belong to the genus *Cirrhipathes* (Antipathidae) and are taxonomically placed within the class Anthozoa, Subclass Hexacorallia and order Antipatharia. These organisms inhabit significantly the reef ecosystem of the tropical and subtropical Indo-Pacific Ocean and the Caribbean Sea. Antipatharia is known as thorny or black corals and has been recognized as rich sources of marine natural products and used in ornamental jewellery markets [1]. In the past, antipatharians were important in the treatment of rheumatism, aphrodisiacs, eye conditions, fever, and other ailments. They were also used as coagulants, and powdered black coral combined with other natural substances was used to treat lung conditions, mouth sores, and other ailments [2]. Antipatharians are harvested commercially

around the world such as in Hawaii, Latin America, Caribbean Sea, Mediterranean Sea, Red Sea and throughout Asia including the Indian coast in the name of black and monkey tail collection by bottom trawl method [3–5]. Due to over-exploitation, it is placed under the endangered species list CITES Appendix II and monitored on a global scale (Convention on International Trade in Endangered Species of Wild Fauna and Flora) [6].

Taxonomically, there are 235 antipatharian species belonging to 43 genera and 7 families around the world so far [7, 8]. In India, Kumar et al. [9] reported eight antipatharian species belonging to seven genera, two families and Kumar et al. [10] added seven more species with an updated list of black coral species (15 species, 11 genera, 4 families) from Andaman and Nicobar Islands, India. Laxmilatha et al. [11] reported 5 species of antipatharians from the Pondicherry coast. The present study reports the new distribution records of two species of *Cirrhipathes* (commercially valuable) at the Gulf of Mannar Marine Biosphere Reserved (GOMBR) as well as the mainland reef ecosystem of India.

The present study was carried out at the Gulf of Mannar Marine Biosphere Reserve, situated at South-east coast of India, from 2017 to 2019 with the help of SCUBA and photographed from the Gulf of Mannar coast. Also, specimens

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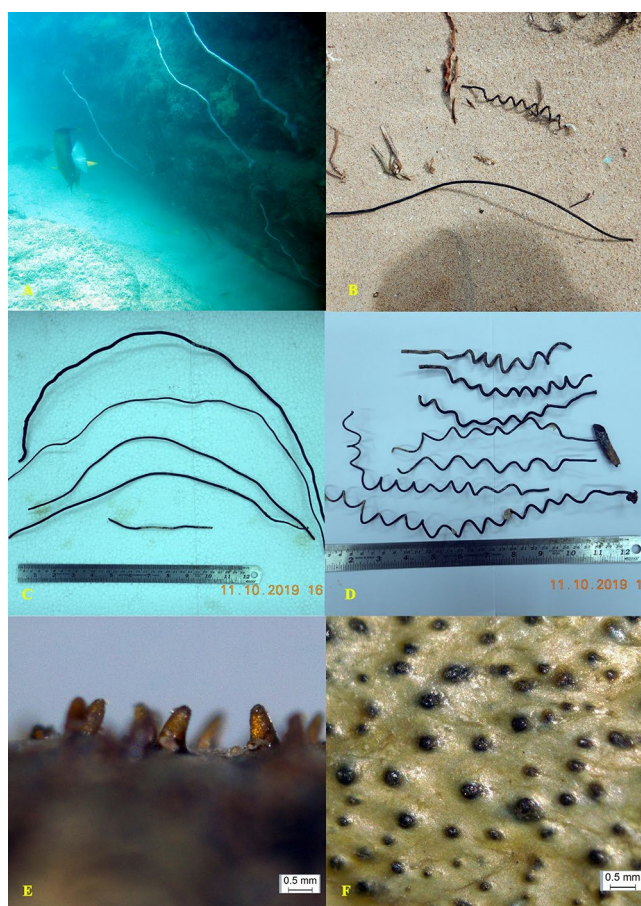


Fig. 1 *C. anguina* (A– Underwater image from Keelakari coast, C– Preserved Sample, E– Microscopic image); *C. spiralis* (D– Preserved Sample, F– Microscopic image); B– *C. anguina* and *C. spiralis* from Pomban coast

were collected from trawler trash in the Pamban fish landing. The collected specimens were identified based on the structure, and morphological characteristics following Opresko [12]. The identified specimens were deposited in the National Zoological Collections of Zoological Survey of India, Marine Aquarium and Regional Centre, Digha, West Bengal.

In India, a total of 16 species belonging to the order Antipatharia were reported. Among them, two species *Cirripathes spiralis*, *C. anguina* are identified and reported as new distribution records from the Gulf of Mannar, Tamil Nadu Coast. The taxonomical identification was made based on the morphological features given below.

Systematic position

Phylum: Cnidaria

Class: Anthozoa

Subclass: Hexacorallia

Order: Antipatharia

Family: Antipathidae

Genus: *Cirripathes*

Cirripathes spiralis (Linnaeus, 1758)

Materials Examined: ZSI/MARC-C6632. A total of 7 fragments were collected on 12.09.2019 Pamban coast, south side (09°16.526 N; 79°12.292 E), Gulf of Mannar, Ramanathapuram, Tamil Nadu.

Description: The specimens were collected from trash materials by trawl fishing along the Pamban shore. The average sizes of the specimens are 15.7 cm to 30 cm in length, unbranched, spring shaped and black-coloured; although live colouration is variable. The axis size is 0.6 mm to 3 mm. Coils are not regular. Spines are cylindrical, flat at frontal part and located at the external corallum side. Spines at the external corallum side are of size up to 0.4 mm while the internal corallum side contains far smaller spines with size up to 0.15 mm. Spines are arranged in eight vertical rows with variable interspacing. Polys radially placed with variable interspacing and absent at the internal side of the coils with a transverse size maximum of upto 1.7 mm (Fig. 1).

Distribution: Worldwide - Sri Lanka, Reunion, New Caledonia; India - Andaman and Nicobar Islands and Puducherry coast.

Remarks: New record to Gulf of Mannar and Tamil Nadu coast. This species is the type species of the genus *Cirripathes*. The description of the species made by Pallas [13] provided the distribution range from Norway, to the Mediterranean Sea except the Indian Sea. Detailed description of the species were done by Brook [14] and van Pesch [15]. The present description coincides with that of van Pesch [15]. The distinguishing trait of this species is its sinistrorse, spirally shaped colony.

Cirripathes anguina (Dana, 1846)

Materials Examined: ZSI/MARC-C6163, one specimen collected on (29.01.2019) from off Chinna Erwadi (08°58.316 N; 78°46.443 E), Keelakari coast and ZSI/MARC-C6633, 6 fragments collected on 12.09.2019 from Pamban coast, south side (09°16.526 N; 79°12.292 E), Gulf of Mannar, Ramanathapuram, Tamil Nadu.

Description: The live animals are unbranched coral whip-like structures and white. The preserved specimens are black. The average height of the live animal is 50 to 200 cm, with axis sizes 2 mm to 6 mm and 0.06 to 0.09 mm spines. Corallum showed a straight appearance in case of smaller colonies, while in larger colonies structure is somewhat sinusoidal. Corallum diameter ranged between 1.2 mm and 5.6 mm. Polyps are creamy white in colour, with transverse diameter of about 0.5 mm to 2 mm and arranged in multiple irregular rows in live colonies. Adjoining polyps are arranged in 0.92–2.1 mm distance from each other. Skeletal spines are conical shaped and some spines show a bifurcation at the frontal part; arranged regularly in horizontal rows. Spine spacing is highly variable (Fig. 1).

Distribution: Worldwide- Fiji Islands, Red Sea, Indo-Pacific; India- Andaman and Nicobar Islands.

Remarks: New record to Gulf of Mannar coast. *C. anguina* can be distinguished from its congenital species by the presence of yellowish tentacles with a brownish base and by the presence of sub-acute laterally compressed spines [9, 10]. The defining feature of this species is the formation of irregularly twisted colonies with spiral coils.

Antipatharia black corals can be seen rarely and very little is known about the basic biology and ecology as most of the species present below 50-meter depth in reef ecosystems. In India, a total of 16 species was reported under this order, among them, *C. anguina*, *C. contorta* were reported from the Andaman and Nicobar coast and *C. spiralis* was reported from the Puducherry coast. The species *Cirrhipathes spiralis* was reported in Galle, Sri Lanka coast during a pearl oyster survey by Herdman [16]. Based on the available literature both species *C. spiralis*, *C. anguina* are the first distribution reports from the Gulf of Mannar and Tamil Nadu coast, India.

Antipatharians have been used by humans in ancient times for the treatment of different diseases and jewellery making purpose. The antipatharian powder was used in wounds before and after cranial surgery in Greco-Roman and the powder was mixed with some natural ingredients applied for mouth sores and also used in lung diseases in Hawaii [17]. Also used to increase virility, cure rheumatism, aphrodisiac and recover eye diseases in Indonesia [18]. Black corals play a major role in recovering pain, and fever, and softening hard masses [2].

The Gulf of Mannar, established as Southeast Asia's marine biosphere reserve in 1989, encompasses a vast expanse of 10,500 square kilometres, including the surrounding coasts and small islands. This region, known as GoM, boasts an extraordinary biodiversity, hosting 3,600 species of marine flora and fauna, making it India's most ecologically diverse marine area [19]. Muthuraman et al. [20] meticulously documented the marine fauna found here, and identified 90 coral species, all of which are protected under the Indian Wildlife Protection Act and these numbers increased to 92 with this present record. However, GoM faces a pressing threat from destructive fishing practices like dynamite fishing, sea turtle poaching, harmful nets, and bottom trawling, which endanger these fragile organisms. This present study is the first to report the presence of these valuable species in the Gulf of Mannar, along the Tamil Nadu coast in India. This information is essential for conservation efforts and better planning. Additionally, occasional discoveries of these precious creatures washed up on the seashore underscore the detrimental impact of fishing activities mentioned above, particularly in areas with coral diversity marked for preservation. To ensure the sustainable

future of these lesser-known species, strict regulation of fishing activities is imperative, particularly in regions where coral diversity is significant and extends to depths of up to 50 m.

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Author Contributions Author JSYK collected the specimen, confirmed the identification and prepared the manuscript. Authors AS, PP assisted in the survey and identification. Author CV and CR coordinated the project and finalized the manuscript.

Declarations

Competing Interests The authors have no competing interests to declare that are relevant to the content of this article.

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